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Designing a choral singing intervention for a psychological study: How to tailor the features of a health-promoting choral singing course for elderly using design thinking

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ABSTRACT

In this method article, we aimed to demonstrate how design thinking can be used as a conceptual framework for planning a scientific, health-promoting intervention for a non-clinical group, with the intention of making it an enjoyable and meaningful experience that the participants are motivated to pursue. The intervention was an 11-month choral singing course specifically designed for healthy participants between the ages 65–75 years. Before launching the intervention, we conducted a preparatory study, which constitutes the current study. A focus group session and a co-creation workshop was held with participants with similar characteristics to those recruited for the upcoming intervention. The findings are presented in an affinity diagram, including an exctract of key aspects identified as crucial for a health-promoting, positive experience, such as the conductor's leadership qualities, motivation, equipment, and more. Based on the findings, we constructed a design specification establishing the intervention content. A participant journey map was developed to consider various aspects of the participants' experiences throughout the upcoming intervention. Our conclusion is that utilizing the design thinking framework for planning health-promoting interventions of psychological studies offers researchers valuable insights and a systemic approach to design the experience to fit the specific needs of the target group.

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Design thinking; health psychology; intervention design; participant journey map; choral singing

Introduction

Psychological health interventions using music are aimed at a variety of target groups and purposes, and require thoughtful planning in order to be

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successful. Some interventions in psychological studies have a health-promoting aim, focusing on disease-prevention (see for example Johnson et al. 2020; Sheppard and Broughton 2020; Viola et al. 2024). Other interventions are aimed at clinical groups, such as rehabilitation after a stroke (see for example Liu et al. 2022; Huang et al. 2021) which poses specific needs for the target group. While it is often natural to be sensitive to the unique needs, constraints and motivations of clinical groups, health-promoting interventions for non-clinical groups are not always given the same care. For clinical groups, it is understood that the participants' experience affects the outcome and the completion rate (see for example Lie et al. 2017; O'Keeffe et al. 2019). For non-clinical groups, researchers are often content to choose an activity or intervention, in our case a choir, and leave important sub-factors (e.g. choir leader characteristics, rehearsal structure, etc.) to chance, when these subfactors in fact may moderate the outcome, or leave unfulfilled potential for a larger effect unexplored. We argue that health-promoting interventions for non-clinical groups too needs to be designed carefully in order to be successful and enjoyable. Few frameworks are available to researchers to conduct preparatory studies that can guide them in the decision-making process when designing health-promoting interventions addressing a specific target group with unique needs, considering there is an infinite set of potential ways of designing it. In the current method article, we propose a novel and flexible approach to how researchers can plan psychological health-promoting activities using the design thinking framework.

Design thinking's increased popularity

Design thinking is a framework or approach that has been applied to a diverse range of domains and disciplines, although it originates from the industrial design field. In recent years, design thinking is commonly understood as the set of practices, methods and approaches used by industrial designers for complex problem-solving. Since its emanation in the 1950s, the approach has gained ground across other disciplines far outside the original domain, such as management, engineering, and health care. Design thinking, unlike other problem-solving approaches, usually deal with problems that are ill-defined, also known as *wicked* (see for example Buchanan 1992; Rittel and Webber 1973; Rylander 2009), and where decisions need to be made based on incomplete or limited information. Furthermore, it encourages out-of-box thinking and generates creative solutions by not limiting itself to a fix set of rules, definitions, or methods, yet employing a systemic view and using measurable outcomes for evaluation.

Design thinking applied to a health-promoting intervention study

Despite the approach's increased popularity within health sciences, and in particular the use of tech-based interventions and education (see for example

Forstag and Cuff 2018; Scholten and Granic 2019), design thinking has not been used much in the field of *planning activities* used for psychological- and/or health-promoting intervention studies. Greater clarity in the link between music, health and ageing has been requested (Creech et al. 2023), as theoretical explanations have so far been limited in their capacity of explaining why and how creative musical processes in groups promote health. In this method article, we demonstrate how design thinking was used to develop a health-promoting choral singing intervention *(Choral singing and healthy ageing*, CSHA) in Sweden, by conducting a *Design research preparation* study, DRP, (see Figure 1). The DRP study was carried out in a separate sample while the CSHA participants underwent an initial passive control period (Figure 1). In order to evaluate the participants' perceived quality and satisfaction of the intervention, an *Evaluation study* (ES), was planned after the completion of the CSHA study. Describing the details of the ES is however beyond the scope of this article.

The CSHA research project

It is well established in the literature that engagement in choral singing carries the potential to promote health and wellbeing in various respects (Clark and Harding 2012; Fancourt and Finn 2019; Herholz and Zatorre 2012). However, the results are mixed, which is not surprising given that there is great diversity among choral practices and target groups. The CSHA was a longitudinal intervention study with the aim of examining the relationship between choral singing and healthy ageing in a within subject design. The study encompassed three data collections, T1-T3 (see Figure 1). The CSHA

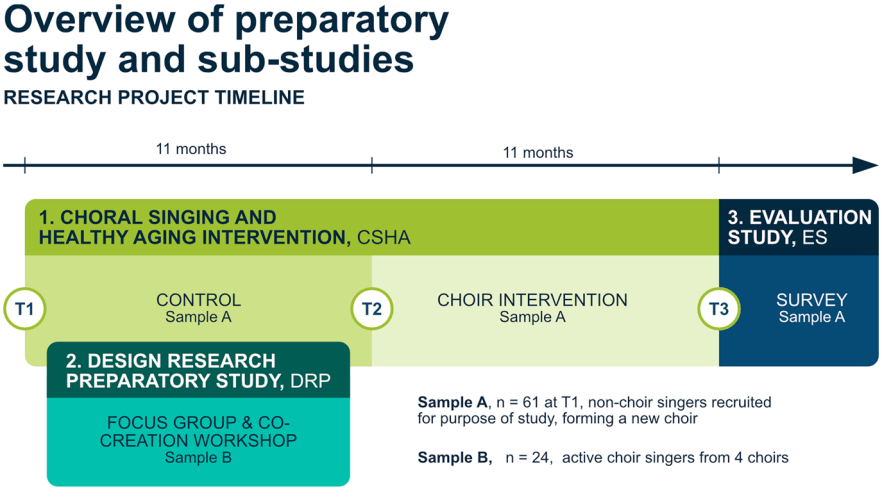


Figure 1. Overview of studies related to the choral singing intervention, where T1–T3 are identical data collections containing various health related measures.

contained a large number and variety of psychological health measures, including: (1) neuropsychological tests, (2) questionnaires on subjective well-being, sleep quality, perceived stress, anxiety, and social relations, and (3) magnetic resonance imaging (MRI) to monitor structural changes in the participants' brain over the different phases of the study and to assess changes in function of episodic and working memory. 61 healthy participants between ages 65 and 75 were recruited to sing in a 11-month choral intervention with weekly rehearsals. During the initial 11-month period between the first and second data collection (T1-T2), no intervention was administered to the participants, establishing this period as the control condition. Following T2, the participants would engage in weekly choral singing for an additional 11 months, (excluding a summer break and public holidays), leading up to the third data collection (T3). Participants were required not to have any choral singing experience in the last 10 years prior to the study.

Design thinking and its possible applications

Design thinking is used extensively in health care in order to improve health interventions, design patient flows, experiences and services linked to care, and for designing medical equipment that needs user-friendliness (Altman, Huang, and Breland 2018; Huang et al. 2018; Leary et al. 2022; Thomas et al. 2020). The closely related concept of *user-centered design* (see e.g. Kling 1977; Mao et al. 2005; Norman 1988), is often used synonymously to, or in conjunction with design thinking. User research, or *design research* is an integral part of the *design process* (see for instance Müller 2021).

We argue that designing a health-promoting intervention study is a wicked problem, because the *problem framing* is a part of the *problem solving,* and there is an infinite number of potential designs of such an intervention. In choral singing, several of the health-promoting elements are interlinked. For example, the social as well as the cognitive aspects of singing in a choir are hypothesized to be causing the potential health benefits, and it is a design choice whether to emphasize one health-promoting aspect over another, or to optimize a combination of both in case there are conflicting health goals. Maximizing cognitive stimulation for memory perseverance (e.g. very challenging repertoire, introducing new repertoire at high pace) might, for example, come at the expense of making the activity less enjoyable and leave less room for socializing, which in turn might lead to drop-out. On the other hand, focusing too much on other aspects than the musical training might decrease motivation as well. In addition, we as researchers also need to choose whether the purpose of the intervention is to *maximize health each promoting element* and create specific guidelines for these, or whether we instead want to *maximize representativity* and *generalisability* of the scientific results by designing a choir that corresponds to the effect of joining a

'regular' choir with no explicit health promotion purpose. Fine-tuning the parameters of the choral singing intervention suggests that the problem of designing an intervention is a wicked one, because there is a co-evolution of problem definition and problem-solving.

Before launching a choral singing intervention, conscious choices need to be made regarding practical issues: for instance, the conductor's characteristics, time and duration for the rehearsals, rehearsal room qualities, and concert practices. Customs differs greatly between choirs, and there is an infinite number of potential intervention designs. A preparatory study can reveal new aspects that could not have been anticipated. For example, the target group consisted of elderly beginners, who might have specific needs regarding availability, comfort, hearing aids, and vocal range. It is conceivable that in the realm of all potential choir interventions, a number of them would be ill-designed and even generate a negative experience. A non-enjoyable experience would neither be representative of choral singing in general, nor would it be suitable for evaluating choral singing as a potential health promoting activity.

Design thinking is an abductive and flexible approach where mixed-methods are often used with the aim of obtaining *actionable knowledge*, i.e. guidance to make informed choices. Before starting the process of proposing any solutions to this *design problem* (namely designing the CSHA intervention), it was necessary to gain qualitative understanding of the group and their situation that was designed for. This phase is usually called *design research*. Commonly used methods for design research (see for example Baxter, Courage, and Caine 2015) are *co-design methods* (Galvagno and Dalli 2014; Steen, Manschot, and De Koning 2011; Zamenopoulos and Alexiou 2018; Yang and Sung 2016), *interviews*, *focus groups*, and *workshops* (Galske et al. 2024).

One well-established way of analysing qualitative data within design thinking is through *affinity diagrams*, sometimes also referred to as *thematic analysis* (Plain 2007). In short, the method involves sorting and categorizing chunks of useful information, *insights*, into meaningful clusters. The design researcher interprets the needs of the participants, and later formulates these needs into a *design/requirements specification* (see for instance Ehn et al. 2021; Righi and James 2010). Preferably, the needs should be expressed in such a way that they allow for more than one solution; the most obvious solution or the first solution that comes to mind is not always necessarily the best one.

Another commonly used tool is the *patient journey map (*see for example Simonse, Albayrak, and Starre 2019), or a *user journey map* (see Endmann and Keßner 2016), which helps the researcher address specific needs that may arise at certain situations. We here introduce the *participant journey map*, that can be used when designing the intervention to identify and mitigate potential physical or mental barriers to completing or fully engaging in treatment.

Aim and research question

The central question guiding this investigation was: How can design thinking be used as a conceptual framework for planning a health-promoting choir intervention for elderly beginners as a part of a psychological study, with the intention of making it an enjoyable and meaningful experience that the participants are motivated to pursue? Our process and usage of design thinking was demonstrated by reporting the method and condensed results for the DRP study that was executed. The purpose of the DRP study was to identify needs of the target group and transform them into actionable information on how to design the CSHA intervention, that would measure the potential health benefits of choral singing in community-dwelling, healthy, elderly participants.

Method

DRP participants

The DRP study consisted of a focus group session and a co-creation workshop held by the first author, both of which were carried out in the same occasion, lasting for about 1.5 hours per group. An information email was sent to 9 choir conductors of Swedish choirs consisting of approximately 20–60 members. Four responded with interest. An appointment was scheduled for each choir, and those interested were invited to join.

A total of 24 participants (18 women) attended. There were four groups in total, and all participants were retired from work (recommended retirement age in Sweden: 67 years), although only 3 out of 4 choirs were explicitly targeting senior citizens. All four choirs explicitly stated on their webpages that they had no audition requirement (i.e. they were open to beginners).

Part I: DRP focus group session

The focus group session was conducted as a semi-structured group interview and the questions were generated by the authors (see Appendix A). The questions were printed on cards and placed in the order they were intended to be answered. Each card was either a *question*, *'finish the sentence'* or a *statements*. Examples of topics were *features of the rehearsal room*, *social atmosphere and activities*, *length and practices of rehearsal and concerts*, as well as *home practice*. Participants were instructed to take the top card from the pile and read it aloud. The reader would answer first, and then the rest of the group would answer as well and discuss it further. The aim of this procedure was to encourage all participants to engage in discussion. Follow-up questions were asked by the interviewer. Each focus group session was audio recorded and a condensed transcription was made by the first author, which were later used for the affinity diagram.

Part II: DRP Co-creation workshop

After the focus group session, participants were asked to form smaller groups of 2–4 individuals. They were given two templates: one for designing their ideal choir, another one for describing desirable as well as undesirable traits and behaviours of a choir leader (see Figure 2). The two templates served as a co-creation medium in the sense that the participants were put in the role of being the co-creators of the upcoming intervention, on the merit of them being representatives of the target group's experiences. As their collective input was directly used by the authors when creating a design specification (Appendix A) for the intervention, the workshop was considered an act of co-creation.

The ideal choir template had 7 different themes: *rehearsal*, *individual practice (between rehearsals)*, *concerts, size of the choir, rehearsal rooms*, *social*, and *repertoire*. Each theme had both some sticky-notes that were pre-filled and some that were blank. Participants were asked to remove the sticky-notes they disagreed with, keep the ones they agreed with, and encouraged to write their own suggestions on the blank ones. At the bottom right, the template also included two themes of reflections on *what it is with choral singing that makes it meaningful* and *what would make you choose one choir over another*. The participants were encouraged to express important preferences that was not covered by researchers' questions.

The template for the ideal conductor was simply an illustration of a conductor. This illustration served the purpose of being a mediating object that would stimulate the discussion. The template contained blank sticky-notes with 2 different colours to describe the desirable and undesirable behaviours and characteristics respectively. One of the groups ran out of time and did not fill out the conductor template.

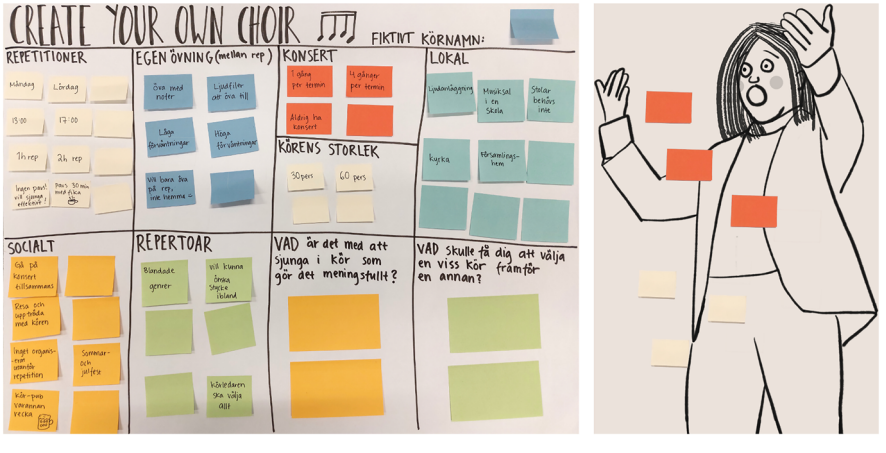


Figure 2. Co-creation workshop material. Left: Ideal choir template, right: ideal conductor template. The ideal conductor template contained blank sticky-notes with 2 different colours for describing desirable and undesirable behaviours and characteristics respectively.

The results of the *co-creation templates* consisted of results from 9 small groups. This content was manifest and verbatim, and not arranged in any cluster. Several factors were found important to consider jointly, for example, one group wanted to practice for 2 hours, and be served coffee for 20 minutes. Another group wanted no coffee and efficient practice for 1 hour. Separating these 2 pieces of information from each other would result in a loss of meaning and undesired reductionism. The results from the co-creation templates were used and interpretated by the researchers along with the affinity diagram when creating the design specification and participant journey map.

DRP affinity diagram

From the condensed transcripts, all relevant data were extracted that answered the research question 'How can a health-promoting choir intervention for elderly beginners be designed, in order to make it an enjoyable and meaningful experience that the participants are motivated to pursue?'. The quotes were coded as an insight: a concise description of actionable information, behaviour, or attitudes. Each insight was written out on a sticky-note.

The insights were categorized by the researchers through the *pain-pleasure framework* (Hashim et al. 2021). Each insight was labelled as one or more of the following categories: *pain* (may contribute to a negative experience), *delights* (contributes to a positive experience), *drives* (motivation), *information* (relevant to understanding the context) or *interviewer reflection* (a new, unanswered question that had arisen).

An *affinity diagram* (see Figure 4) was then constructed by the first author from the insights, sorted into non-pre-determined categories (clusters): one insight at a time was taken and placed in either an existing cluster or a new one was created. Once all insights were sorted, they were further sorted into smaller clusters if necessary. This was an iterative (abductive) process in which insights and clusters was re-sorted and re-formulated as many times as needed.

CSHA participant journey map

A *participant journey map* was developed by the authors in parallel with the analysis of the material from the DRP study. The aim was to tailor the participants' experience in the CSHA's journey through the entire research project: from advertisement of the study to the evaluation study (ES).

CSHA design specification

A *design specification* was made based on the insights from the DRP study (Appendix A). The design specification was a type of blueprint for the upcoming CSHA intervention, compiling a list of needs and expectations the CSHA

intervention should aim to meet. The design specification could also be interpreted as intervention guidelines or a manual.

Design process

Figure 3 visualizes the design process of the CSHA intervention using the double diamond model (Banathy 2013; Design Council 2013). At the beginning of the process (1), the DRP study was initiated by preparing questions for the target group that would explore the problem from an open perspective that would allow for several problem-framings, yet generate actionable knowledge. In this exploratory phase, divergence of ideas and questions would be sought, in order to get as broad a perspective as possible.

In the next step (2), we would try to understand and synthesize the findings from the DRP study, and translate it into a participant journey map and a design specification. This phase would be characterized by a convergent approach where we tried to define, based on the insights, the type of intervention most aligned with our CSHA research question: How does choir singing promote emotional well-being, social engagement, and cognitive reserve in healthy elderly people?

In the upcoming phase, we implemented our findings (3), by, for example, recruiting a choir leader, looking for a rehearsal room, and define the structure of rehearsals and information regarding expected home practice. Furthermore, equipment was purchased and planned in greater detail. This phase would also be characterized by divergence, i.e. by generating ideas and aiming for several different solutions that we could then choose from.

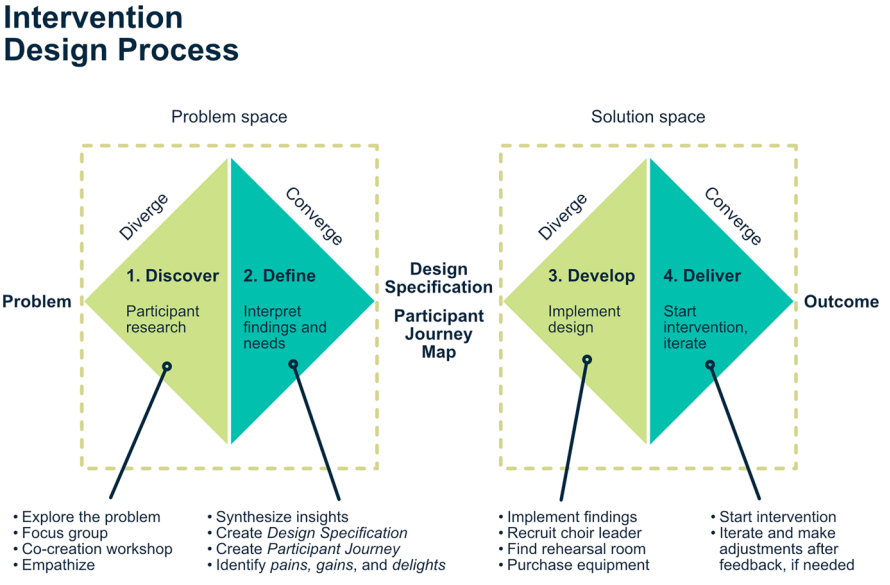


Figure 3. Intervention design process.

For example, one solution could be a rehearsal of 2 hours, including a 30 min break. Another solution could be a rehearsal lasting for 1.5 hours, including a shorter break.

Lastly, once the rehearsals would start in the CSHA intervention (4), an iterative approach would be applied, and feedback from the participants should be considered continuously in order to ensure a positive experience throughout the intervention. For example, the need for additional purchases would be identified, as well as discoveries of new problems that would not or could not have been identified beforehand.

Ethical considerations for preparatory study

Before starting the DRP study, written and informed consent was obtained from all participants. The study was reviewed by the Ethics Committee of South East Sweden (diary no: 836-2022). Data were handled according to GDPR and Swedish legislation (SFS 2003:460), as well as the Declaration of Helsinki (World Medical Association 2013). The transcripts were confidential and pseudonymized.

Results

Affinity diagram clusters

Figure 4 presents the themes that emerged from the affinity analysis. Several of the themes were related to each other, as indicated by the dashed nodes.

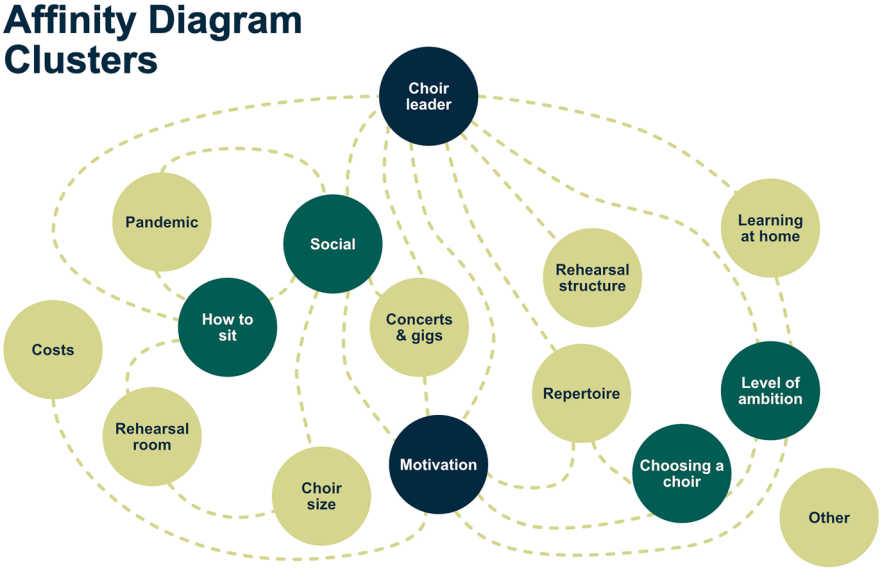


Figure 4. Affinity diagram clusters.

In this section we present some condensed examples of the key insights. The main purpose of the current results section was to illustrate how the design thinking framework can be applied to intervention research, and communicating results from design research may benefit from being presented in more detail.

When creating the design specification, it was necessary to consider several aspects simultaneously; for example, it was found that the social experience as a whole of singing in a choir was influenced by several aspects: the rehearsal room's capacity to change the positions of the chairs, the conductor's leadership, the community that arises at concerts, and the size of the choir. The affinity diagram provided the research group with a systemic understanding of the context and participants. The themes were colour coded according to how many nodes each theme had (i.e. connection to other themes). As can be seen in Figure 4, *Choir leader* and *Motivation* were the two themes that were connected to most other themes, followed by *Social*, *How to sit*, *Choosing a choir*, and *Level of ambition*.

Choir leader

The choir leader played a key role in the satisfaction of the choir members and needed good leadership skills to create a positive atmosphere, for example by providing positive feedback to the singers, being flexible, and patient. In addition to these qualities, the conductor should also have musical skills in singing, accompanying by piano, conducting, etc. Many choristers expected to be taught some vocal technique and do intonation exercises in addition to rehearsing the repertoire. Three recurrently mentioned undesirable conductor behaviours were: singling out members who make mistakes, making rehearsal run over time, or being unstructured in general. Choir singers tended to leave a choir if they did not like the conductor. Many participants also expected to receive learning tracks from the conductor to practice along with at home. Furthermore, the choir leader often had an influence on most aspects of the rehearsal structure: what and how to practice during rehearsals as well as the time management, or providing shorter breaks, for example.

Motivation

Motivation, in our context, pertains to the factors that drive participants to engage in regular choir singing. There were several recurring motivations for singing in a choir; in the DRP study, 2 motivations in particular recurred: (1) *social reasons*, and (2) the *desire to musically express oneself by singing*. For some, one reason was more important than the other; for some, both aspects were equally important. Furthermore, many mentioned being able to express themselves musically together *with other voices*. Several participants reported that singing in a choir gave them energy, made them feel relaxed, and they

enjoyed doing it. Some mentioned that they had a belief that choral singing and music were healing and health promoting. Being able to sing with grandchildren, keeping the voice in good condition, practicing posture and breathing were also mentioned. For the sake of motivation, it was also important to have some kind of concert or performance every semester to look forward to. Furthermore, it was important that the choir met the expectations and wishes of individual choir members. More on this can be found in the section on '*Choosing a choir*'.

Social

The social aspects of singing in a choir were largely influenced by the conductor's ability to create a positive atmosphere, and by using strategies to get choir members to get to know each other and achieve social cohesion. Similarly, the characteristics of the rehearsal location shaped the social environment, such as whether there was a natural place to meet before or after rehearsal to socialize. There were also organizational aspects, such as the planning of get-togethers in connection with a concert or the end of the semester. Since choir members were expected to be silent during rehearsals unless they were singing, it was precisely the occasions *outside rehearsal* that allowed members to get to know each other. An important recurring aspect was the experience of being new to the choir and whether or not being included and taken care of.

How to sit & rehearsal room

How the members sat turned out to also have a social aspect. For example, a new member would feel insecure about where to sit and whether taking someone else's seat. There were both advantages and disadvantages to fixed seating, but regardless, there was a desire for clarity about it. Most participants preferred sitting in the shape of an arc to be able to hear and see each other and the conductor. The rehearsal room could also limit positioning of the seats: benches or cinema seats cannot be moved into an arc. Similarly, the comfort of the seats was an important aspect. An aesthetically pleasing surrounding and beautiful acoustics also contributed to a positive experience. The equipment used by the choir varied, but accompanying instruments (essential), microphone, amplifier, speakers, and projector were mentioned.

Choosing a choir

It was considered important that the choir matched the needs and expectations of the individual choir singer in order to be enjoyable. One person's pain could be another's delight, and most often, participant sought out a choir whose explicit description met their preferences. For example, some choirs have niched themselves to a particular *genre*, such as gospel or

chamber music, while others sing a mixed repertoire. Furthermore, a choir may target a certain *skill level*: beginner, intermediate, or advanced, and audition requirements are quite common. It was considered necessary that the skill level was neither too difficult nor too easy in relation to one's own ability. A couple of participants said that they preferred to sing with others of different ages and voices, as they got to know different types of people that they might not have got to known otherwise. Likewise, it was reported that singing in large choirs (100+ members) was a powerful musical experience, but at the expense of social community, which was considered easier to create in a smaller choir. Furthermore, some expressed that it could feel intimidating to hear their own voice, while others, on the contrary, found it wonderful to be heard a lot.

Level of ambition

In many choirs, but not all, the members were expected to practice at home between rehearsals. Sometimes there were learning tracks provided by the conductor to support the learning, and it seemed to be appreciated both by those who read and those who did not read music. Furthermore, most participants seemed to find it important that the choir leader had high expectations of the choir's ability to learn in order for them to feel motivated – regardless of skill level. All participants expressed that the repertoire needed to contain more than one voice (i.e. non-unison) to be stimulating.

Design specification

Using the insights gained from the affinity diagram analysis, the research team created a *design specification* for the CSHA intervention (see Appendix A). Usually, it is the choir leader or members of the choir that make some of the decisions that the CSHA research team needed to make. It was therefore considered important to try to find a balance between the findings of the DRP study, and to give the recruited choir leader the autonomy to use their expertise and experience to make qualified decisions. Furthermore, the research team sought to maintain in the CSHA intervention design what is *essential* to choral singing, that is, not to add other potentially health-promoting elements that is an uncommon in choral singing practice. This was an important position, considering that the CSHA study aimed at drawing generalized conclusions of the effect of choral singing on health.

Participant journey map

In parallel with the creation of the design specification, a participant journey map (see Figure 5) was developed after all members of the research team had reviewed the findings. The participant journey map illustrated the

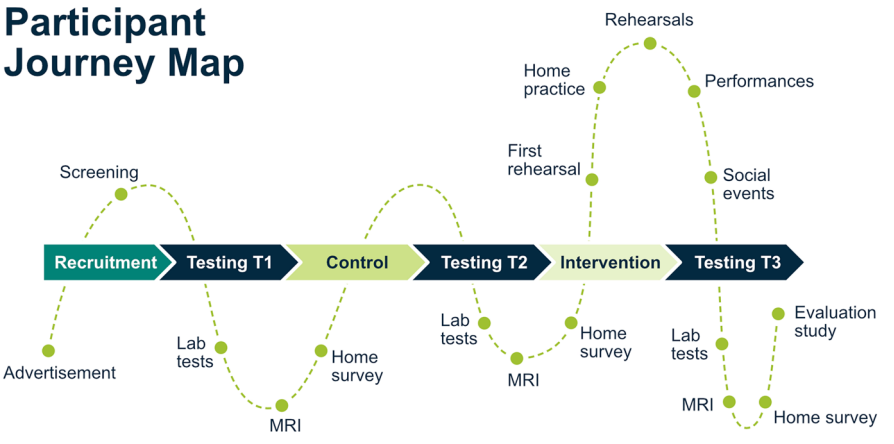


Figure 5. Participant journey map.

participants in the future CSHA intervention's interactions involving all parts of the project: *recruitment*, *testing*, *control period*, and *intervention*. The findings of the DRP study had the greatest influence on the intervention phase, although the function of the participant journey map was to consistently see the participants' perspective and try to maintain their initial motivation to complete the participation. This of course also applied to recruitment and testing, as the research team needed to plan the communication and detail of information the participant needed to receive at what time. A critical phase that was important to plan well was the beginning of the intervention including the first rehearsal, to make sure it provided a positive experience and participants would choose to continue. This included, for example, receiving clear instructions on how to find the rehearsal location, feeling welcome and knowing where to sit, and other important accessibility qualities of the rehearsal room, as well as being allowed to sing and try out the activity at the very first occasion. Some other aspects were considered to be more important to achieve over time, such as creating engaging rehearsals and a good social atmosphere.

Discussion

In this study, we showed how design thinking can be used as a conceptual framework to create a health-promoting choral singing intervention for a psychological study, with the intention of making it an enjoyable and meaningful experience that participants are motivated to pursue. The aim was to gain actionable knowledge concerning the needs of the target group (i.e. healthy, elderly participants who were beginners in choral singing) through a qualitative Design research preparatory (DRP) study, and thus together with the Choral singing and Healthy ageing (CSHA) study and an evaluation study (ES) constitute a mixed-method approach.

Preunderstanding

The research team conducting the DRP and CSHA study was a cross-disciplinary group with professional backgrounds in the fields of industrial design engineering, psychology, neuropsychology, music education, choral conducting, and choral singing. This multiplicity of expertise contributed both to the understanding of the participants' choral singing experience, as well as to important methodological aspects of the study design. Mixed-methods have been suggested to strengthen the evidence base in the field of music therapy (Bradt, Burns, and Creswell 2013). More specifically, design thinking has been suggested to be an approach that could enhance psychological interventions (Yeager et al. 2016).

Implications and significance of the DRP findings

We argue that design thinking is highly useful to researchers in the process of systematically gaining important insights and in the decision-making when planning a health-promoting intervention study. Choral singing is an activity with a complex set of factors (e.g. conductors' leadership, characteristics of the rehearsal room, etc.) that interact in their contribution to the choral singing experience. These factors cannot be considered separately without the loss of an integrated perspective, and design thinking provides a framework to tailor a holistic experience for the target group. Furthermore, using design thinking has also successfully generated actionable knowledge for the researchers. It has provided support for decision-making in the development process of the CSHA intervention, where there was an infinite number of potential solutions, and where decisions needed to be made continuously (participant journey), and at times with incomplete information.

Furthermore, we believe that a practical implication of our DRP study is that it contributed to the decision-making regarding the CSHA intervention and that it increased its quality (e.g. generating an enjoyable singing experience, finding a competent conductor etc.) and ecological validity (intervention in accordance with choir singing practices) that can be generalized to choir singing in general. Likewise, we argue that our explicit reporting of the decision-making process enables the comparison of the CSHA results with similar studies, as we have clearly described the design specification for our intervention as well as the rationale for how we arrived at these decisions.

Results discussion

The results from the focus group session suggested that singers intentionally join a choir that matches their own preferences. For example, someone that prefers gospel might try to join a gospel choir instead of a classical chamber choir. In the CSHA project, no clear profile of the choir was advertised other

than targeting healthy beginners aged 65–75. However, it is likely that the participants of the CSHA project interpreted the absence of information regarding the genre profile that they would, by implication, try out different genres.

It was also noted that choir singers expressed that they would leave a choir if they did not enjoy it. Accordingly, we expect some degree of selection bias in the interviewed sample: all interviewees were presumably happy with their current choir, or they probably would have quit. For obvious reasons, it is difficult to find participants to interview who have left a particular choir. On the other hand, several of the participants had previously sung in other choirs and were therefore able to make comparisons. Furthermore, several important aspects about being a senior citizen and a beginner were identified, which were not anticipated by the authors in advance. We believe these findings were likely discovered due to the semi-structured nature of the focus group interview, allowing new topics to be addressed. An example of such topic was singing in a mixed-aged group verses same-aged group, which was found to be a highly individual preference.

An imperative difference between the DRP study sample and the CSHA sample, is that the CSHA were forming a completely new choir. This meant that the members would not know each other beforehand, and there would be no social culture nor practices to conform to. Instead, the CSHA would have to shape this culture together and with the choir leader. The findings of the DRP study provided the researchers with insights to reflect upon this aspect and try to mitigate potential negative impacts, by for instance planning the first rehearsals in a manner that promoted social interaction. Furthermore, it was noted that most choirs and choir leaders had some degree of autonomy regarding decision-making for the choir. We tried to intentionally leave room for some decisions to be made by the choir members and the choir leader, in order for them appropriate the choir in an organic the way they would have done had it not been an intervention lead by researchers. This attempt, we argue increased ecological validity.

Finally, it is also important to note that a choir consists of group processes where the members develop over time both as a group and as individuals. At the beginning of the intervention, they would have no recent choir experience, but perhaps some expectations of what it would be like. At the and, on the other hand, they have learned new pieces, new musical skills, made new friends, as well as shaped new preferences. We as researchers can only design for an *intended experience* (e.g. good social atmosphere), but of course we cannot control all factors in a group process. This is a known limitation of design thinking – it is only possible to design *for* the experience, not the experience itself, as an experience is something that arises within the participant. Nevertheless, the findings helped us to adopt a reflexive approach in anticipating expectations regarding for example the rehearsals, concerts, and social activities.

Limitations

As in all design problems, there may be practical factors that limit what can be implemented from the design specification into the CSHA intervention: for example, a constrained budget or limited options of rehearsal rooms. However, these are constraints that regular, non-experimental choirs also need to cope with, and should thus not pose a serious problem for the CSHA intervention quality. Practical limitations for implementing certain design features are something that can be included in the analyses and decision-making processes within the design thinking framework, and which we highly recommend when sensible. Sometimes weighting needs is useful, which could be done through a *morphological analysis* (Álvarez and Ritchey 2015; Zwicky 1960). Such analysis could have further supported our process when choosing between different rehearsal rooms, as many needs differ qualitatively. For example, one room may have better acoustics, while another has better opportunities to place the seats into an arc. In our study, we have not weighted the needs, but we recommend doing so if considered necessary.

There are also limitations imposed by the methodology of the CSHA study. For example, the CSHA study examined a sample aged 65–75, which naturally makes it difficult to create an age-mixed group, which was a feature that the DRP study suggested was a frequently occurring preference. The same applied to the size of the choir, which was fixed due to scientific reasons, namely the power analysis made for the MRI studies.

Finally, an integral part of design thinking is proof of concept by *iteration* and *validation* of the design. Several iterations can be difficult to achieve in a research context, which usually provides less flexibility in terms of making changes in ongoing projects. However, we will in the ES measure (1) how well the participants perceived the choral singing intervention met the criteria of the design specification, and (2) the overall satisfaction with the choral singing course measuring the participants *Net Promoter Score* (Reichheld 2003) and *customer satisfaction* (see for instance Hill and Alexander 2017). This study is planned to be carried out after the CSHA intervention has ended.

Future studies

Future health-promoting intervention studies may benefit from using design thinking and mixed-methods to gain qualitative and systematic understanding of their participants and the context of their intervention. We believe that design thinking is useful for creating interventions for psychological studies, as the purpose differs greatly between different interventions and target groups. In our CSHA study, we targeted healthy participants aged 65–75 years, which placed unique demands on, for example, accessibility and the condition of the mature voice. Health-promoting interventions aimed at other target groups and purposes, for example stroke patients or children with autism, naturally would

have placed very different demands on the intervention design. In this regard, we believe that design thinking as a framework is invaluable, as it aims to flexibly adapt the design of the intervention to its target groups and purposes.

Conclusion

We conclude that design thinking has the potential of improving health-promoting interventions specifically, and psychological studies more generally. We have demonstrated this by reporting how we used design thinking in a preparatory study, and later implemented the findings to a larger research intervention study; we showed that a health-promoting choir intervention for elderly beginners can be successfully designed using *design thinking* frameworks and methods such as *focus groups, co-creation workshops*, *affinity diagramming*, *participant journey mapping* and *design specification*. Preparatory studies on a target group can help researchers improve the quality of their intervention by *gaining actionable knowledge* and providing a *systemic approach*. The preparatory study, intervention study, and an evaluation study jointly constitutes an iterative, mixed-method approach.

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Appendix A

Focus group questions

Questions (Q), 'fill in the blanks' (F), and statements (S) used for semi-structured focus group.

English translation of Swedish questions	
F	Our rehearsals usually last for _____ minutes
S	We usually have a break during rehearsal
S	We usually alternate between sitting and standing when we sing
S	When we sing, we usually stand in a semi-circle
Q	Do you usually have fika* during rehearsals?
Q	Does your choir socialize outside rehearsal?
S	In the choir, everyone is at beginner level
Q	Are you expected to practice between rehearsals? If yes, how (sheet music, audio tracks)?
F	When the demands are high, I feel _____
F	When the demands are low, I feel _____
S	Choir singing is an expensive activity
S	It is important to like the choir repertoire
S	It is important to be able to suggest pieces
S	The characteristics of the rehearsal room are important – e.g. beautiful environment, good acoustics, good ventilation
F	I sing in a choir because _____
Q	Does it feel different to sing in a small choir compared to a large one? In what way?

**Fika:* Swedish for having coffee, tea, snack, or something sweet.

Appendix B. Design specification.

Design specification	
1	Choir leader
1.1	Flexibility
1.2	Sense of humour
1.3	Positivity
1.4	Pedagogical ability
1.5	Patience
1.6	Does not single out anyone in a condescending way if they make mistakes
1.7	Have positive expectations on choir's ability to learn
1.8	Skilled vocalist
1.9	Skilled accompanist
1.10	Skilled conductor
1.11	Structured and prepared
1.12	Provides choir with audio files for practice when needed
1.13	Provides choir with sheet music for practice when needed
1.14	Open-minded when a choir member has a suggestion
1.15	Holds good warm-ups
1.16	Trusts choir's ability to perform at a concert
1.17	Sensitive to the fact that singing in front of others could cause performance anxiety
1.18	Provides positive feedback
1.19	Educates the choir in music, breathing, vocal techniques, and posture

(*Continued*)

Appendix B. Continued.

Design specification	
1.20	Does not make rehearsals run late
1.21	Matches the musical content to the skill level of the choir
1.22	Makes use of the choir members' other talents besides singing
1.23	Does intonation exercises
1.24	Chooses a varied repertoire
1.25	Chooses repertoire that the choir is satisfied with
1.26	Chooses repertoire that consists of at least two voices
1.27	Set aside brief moments during rehearsals for socializing
2	Concerts and performances
2.1	Approximately one concert or performance is arranged each semester
2.2	Fika should be served to the choir before or after a performance
2.3	Instructions on how to dress on concerts should be given
3	Rehearsal structure
3.1	Rehearsals should not be scheduled too late at night
3.2	Rehearsal should last for 1h 30 m
3.3	There should be breaks
3.4	The rehearsals should be scheduled weekdays
3.5	The rehearsals should be scheduled daytime
3.6	There should be rehearsals scheduled occasionally for each voice to practice individually
4	Social aspects
4.1	There should be a possibility to fika before, during, or after rehearsal
4.2	There should be a convenient place to sit to fika before, during, after rehearsal
4.3	The atmosphere should encourage spontaneous interaction among the choir members
4.4	The choir members should experience a sense of coherence
4.5	The choir members should experience a positive atmosphere in the choir
4.6	Social activities should be organized at end of semester or after performance
5	Rehearsal room qualities
5.1	The choir should be able to sit in a way so that they hear themselves well when singing
5.2	The choir should be able to sit in a way so that they see each other when singing
5.3	The choir should be able to sit in a way so that they see the conductor
5.4	The choir should be able to sit in a way so that they hear when the conductor speaks
5.5	The seats should be comfortable
5.6	It should be possible to both stand and sit while singing
5.7	The choir members should always be able to know where they can sit
5.8	Should be inspiring
5.9	Should have good acoustic features
5.10	Should be in good condition
5.11	Should have good air / ventilation
5.12	Adequate size of room
5.13	Comfortable temperature
5.14	Be equipped with PA and microphone
5.15	Be equipped with accompanying instruments
6	Size of choir
6.1	Should be small enough to create a sense of community
6.2	Should be small enough to be manageable by the conductor
6.3	Should be large enough for each voice to have confidence also when there are absences